Does the network know what film history knows?

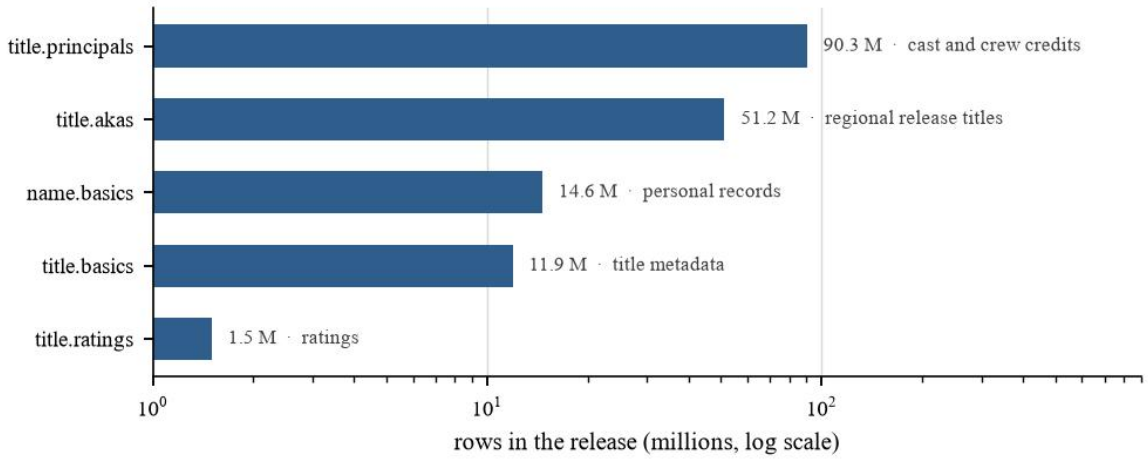
Parallel graph analytics of the Hong Kong release network

CS5488 Data-Intensive Computing · Group project · November 2026

170 M rows scanned · 388,168 co-star edges · three parallel tiers on 20 cores

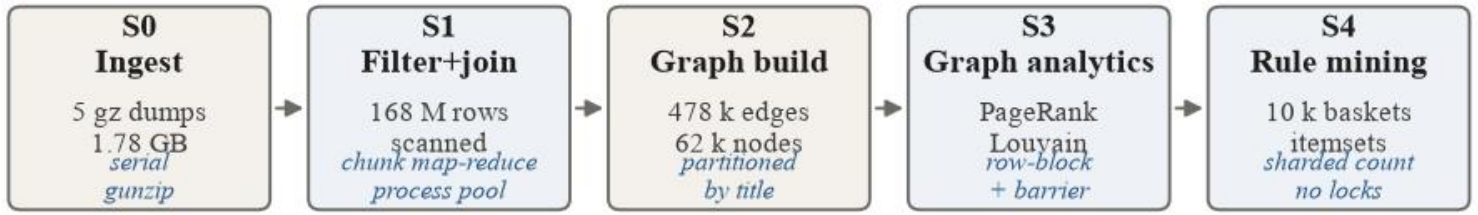
The data has no country field — so we had to choose

- IMDb public dumps: 1.78 GB compressed, 9.2 GB of text, 169.6 M rows
- Five files: titles, release aliases, cast/crew credits, people, ratings
- No production-country column exists in this release
  - Three candidate filters, three different answers:
  - region = HK → 25,997 titles (admits every import)
  - region = HK + Chinese alias → 15,715 (selects the imports)
  - region = HK + Chinese original title → 0 (IMDb romanises)



Decision: keep the release market, state it plainly, and let community detection separate the cinemas. That turned a data defect into the main result.

Three workloads, three parallel strategies



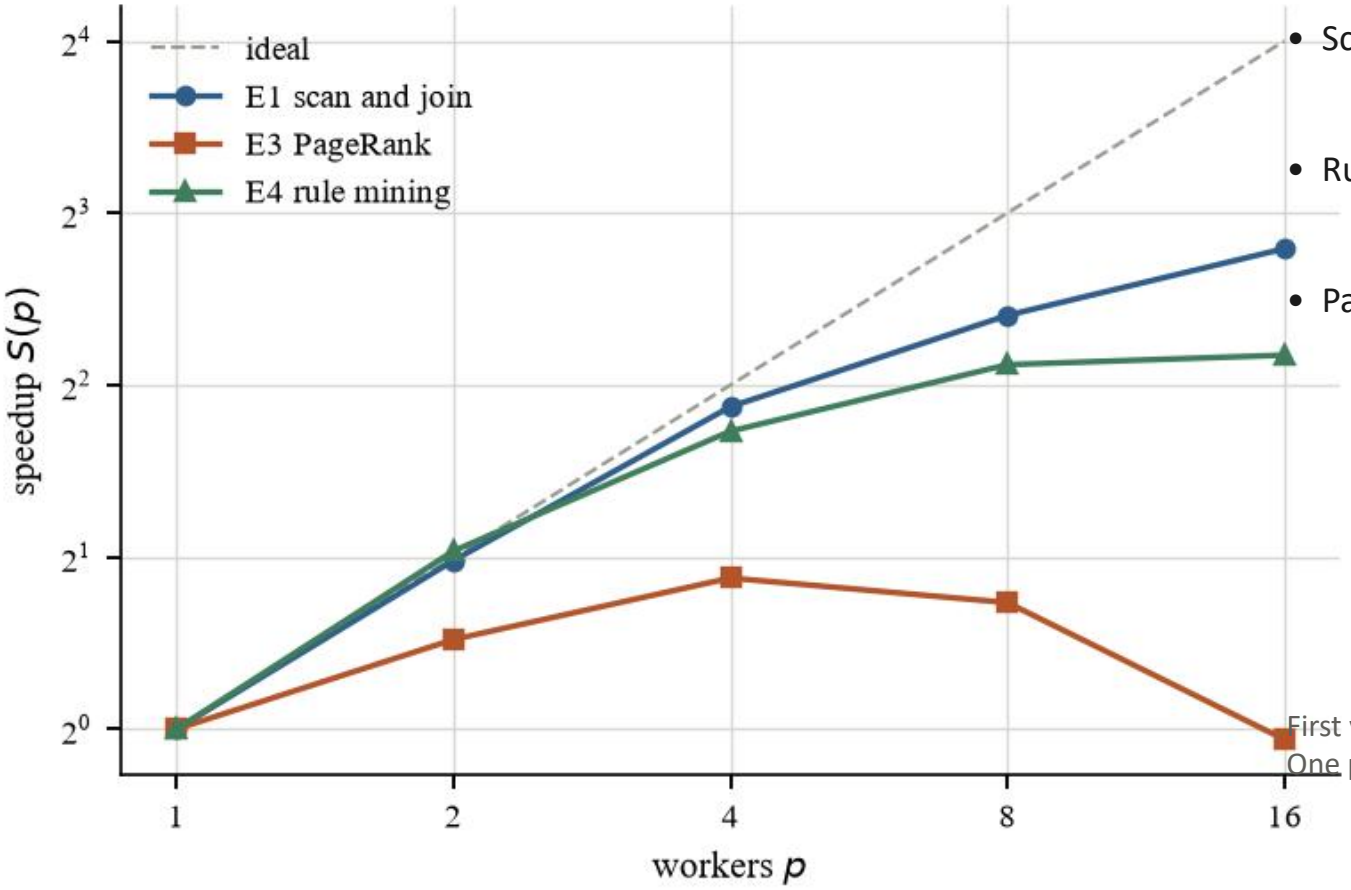
parallel tiers

- ☐ tier 1 chunk map-reduce over byte ranges (S1)
- ☐ tier 2 blocked sparse matvec, barrier per iteration (S3)
- ☐ tier 3 sharded candidate counting, no locks (S4)

- Tier 1 · byte-range map-reduce, process pool
  - Parse-bound → the GIL rules out threads
- Tier 2 · blocked sparse matvec
  - Rows partition; one barrier per iteration
- Tier 3 · sharded counting
  - No locks during the scan; merge once

gzip is not seekable, so decompression is serial by construction (33.8 s, 4% of the serial cost). We report it separately instead of hiding it inside a stage.

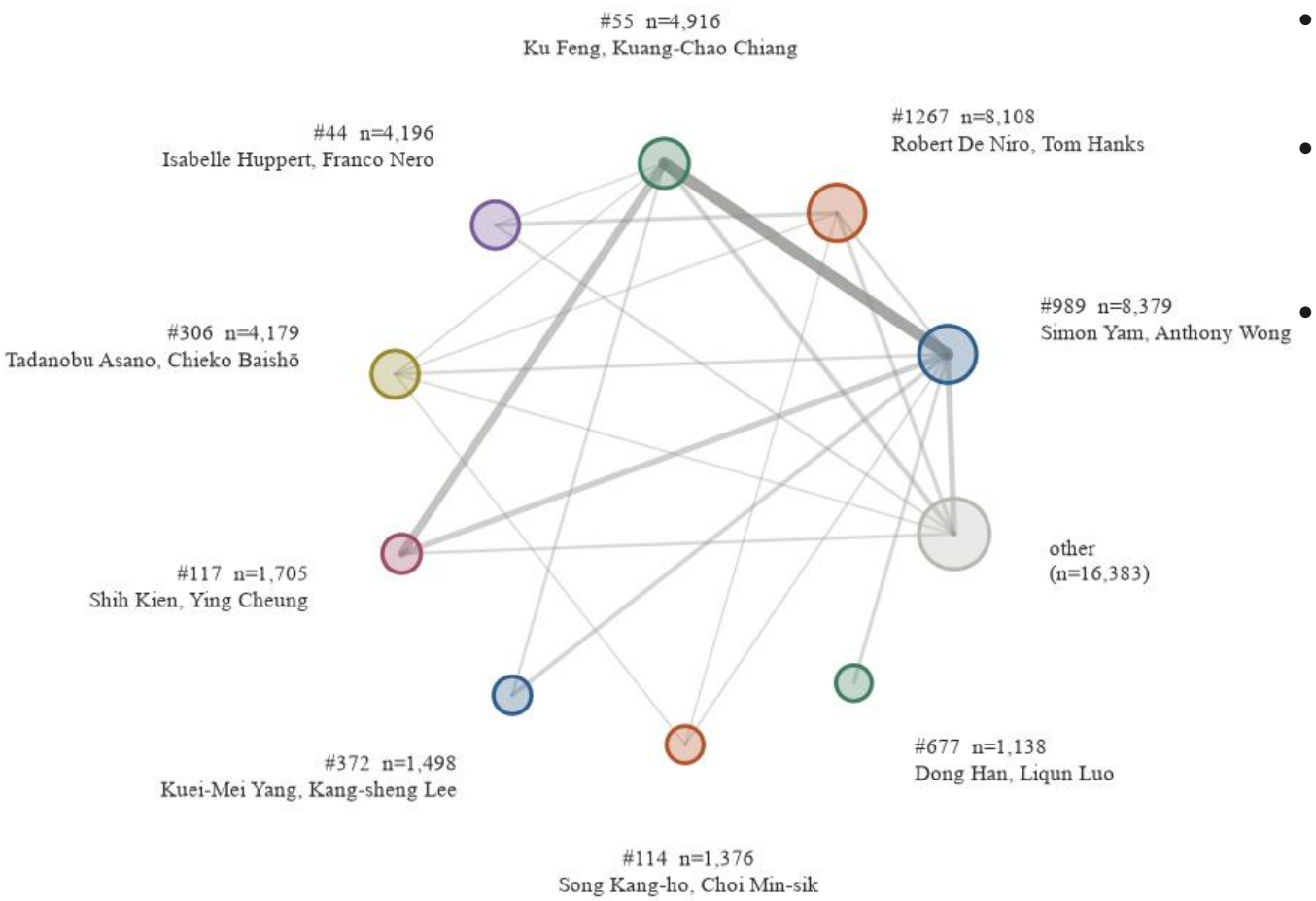
Same machine, three ceilings



- Scan 6.9× at 16 workers
  - serial merge is the tail
- Rule counting 4.5× at 16 workers
  - candidate generation saturates at ~1×
- PageRank 1.0× — and 2.1× at 16× the graph
  - bandwidth-bound, not overhead-bound

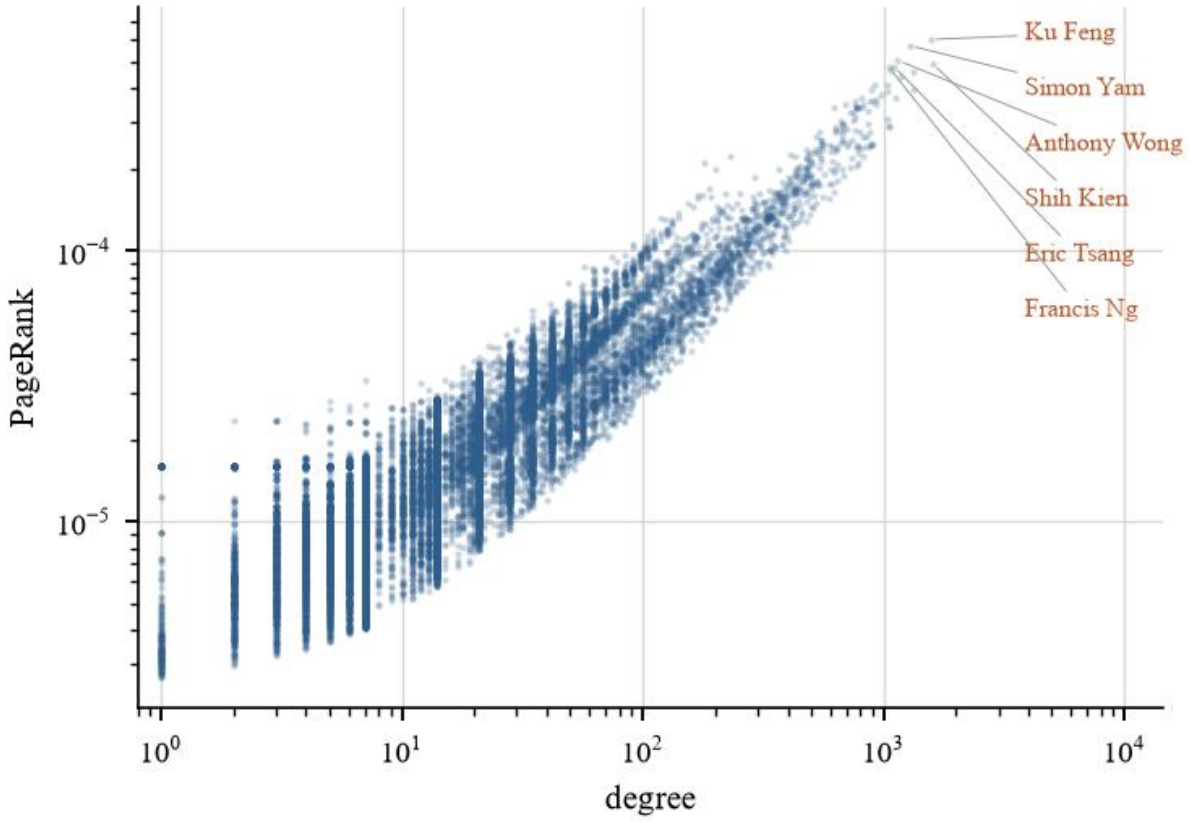
First version created a pool per stage: 2.6× peak, then slower at 16 workers. One pool, one reference file per worker: 6.8×.

The blocs are real, and nobody labelled them



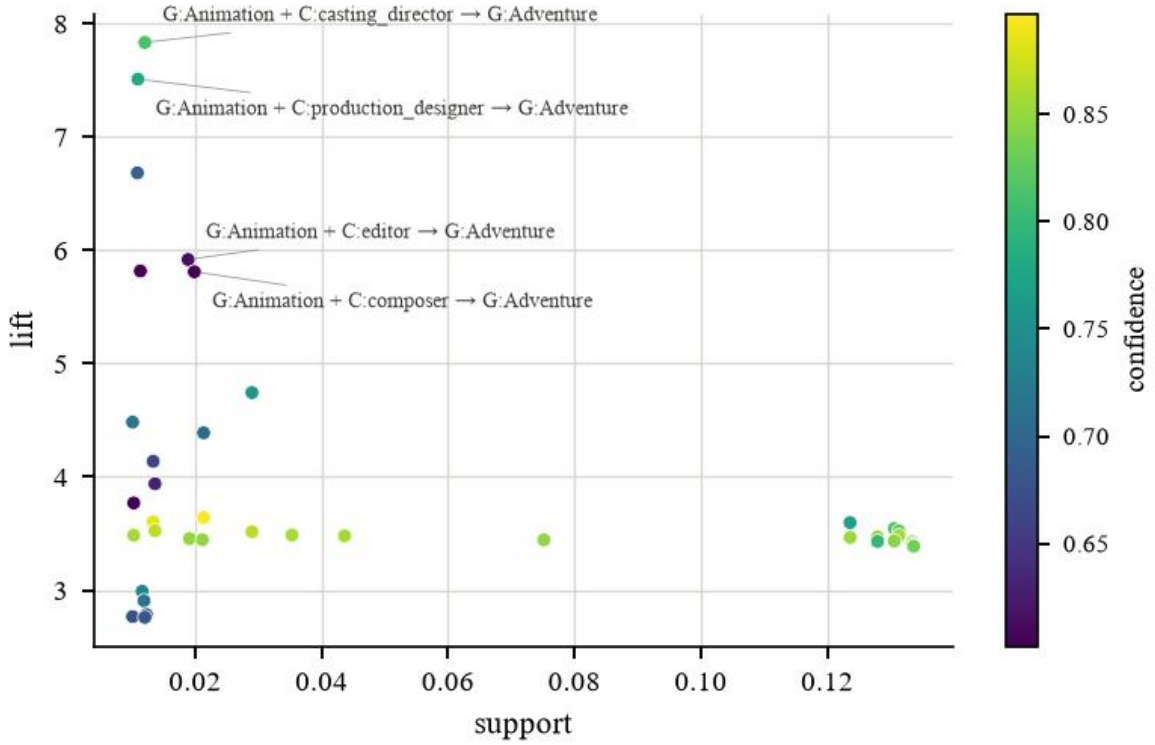
- Louvain: Q = 0.75, 1,285 communities
  - Cross-seed stability NMI = 0.89
- Nine blocs cover 68% of 51,878 people
  - Hollywood · Cantonese · mainland · Shaw-era · European · Japanese
- Label propagation collapses: Q = 0.09, 87% in one label
  - Cheap, and useless on this graph

Inside the local bloc, rank ≠ volume



- PageRank and degree correlate at ρ = 0.53 — related, not the same
- Top rank: Ku Feng, Simon Yam, Anthony Wong, Shih Kien, Eric Tsang, Francis Ng
- These are the people the industry's own histories name
  - Several higher-degree performers do not appear
  - They accumulated credits, not central collaborators
- Decade profile: the local bloc peaks before 1990, the mainland bloc after 2000

Rules validate the pipeline before they explain it



- 652 rules at support ≥ 1%, confidence ≥ 0.6, lift > 1.2
- Strongest are structural: animation → adventure, lift 7.8
- Crew-role pairs above 2% support are the studio signature
  - Technical roles travel together in assembly-line production
  - Weakest after 2000 — the comparison is future work
- Career clustering (k = 8, silhouette 0.66) agrees with the graph ranking

What we would do next, and what we would not

- Replace Apriori with FP-Growth
  - moves the ceiling from the irregular phase to counting
- Get a country label from a dump, not a SPARQL endpoint
  - removes the largest threat to validity
- Dynamic community detection instead of fixed-label flux

- What we would not do again
  - run PageRank in parallel at 10^5 edges
  - spawn a pool per stage
  - trust a release-region filter as a production filter
-
- Reproducible: every number from one command
  - python src/bench.py --exp all